Cameron Emery

Systems, Embedded & AI-Native Product Engineer

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End-to-end Systems, Embedded & AI-Native Product Engineer who takes ambiguous problems from architecture and physical bench bring-up to verified full-stack products and distributed systems at single-engineer velocity. Currently serving as the sole software/systems engineer at Electronic Mobility Controls while building open-source systems across TypeScript/React, Rust, Lean 4 formal verification, and custom LLM architectures, and contributing upstream to **Tenstorrent (`tt-metal`)** and **T3 Code (`t3code`)**.

Technical Capabilities

Languages & Formal Verification: TypeScript, JavaScript, Rust (Stable & `no_std`), C, C++, Python, SQL, x86_64 Assembly, Verilog, Lean 4

Full-Stack, AI Agents & ML: React, Next.js, Node.js, FastAPI, PostgreSQL, SQLite, Prisma, Model Context Protocol (MCP) Servers, PyTorch, SafeTensors, NVFP4

Embedded, Silicon & Buses: NXP S32K344, Lattice iCE40UP5K FPGA, Okika AN231E04 FPAA, i.MX8/i.MX93, CAN-FD, LIN, SAE J2716 SENT, Automotive Ethernet, CXPI

OS, EDA & Manufacturing: Embedded Linux (Yocto/BitBake, Qt/QML), KiCad Schematic/PCB Design, Oscilloscopes/Logic Analyzers, Fusion 360 / FreeCAD, 3D Printing & DFM

Professional Experience & Upstream Open Source

Software & Systems Engineer (Sole Engineer) — Electronic Mobility Controls | Aug 2025 – Present (Augusta, ME)

- **End-to-End Ownership:** Serve as the company's sole software/firmware/systems engineer across safety-critical automotive drive-by-wire accessibility platforms—owning architecture, C/C++ firmware, PCB layout, bench bring-up, vehicle integration, and automated testing.
- **Universal Automotive Interceptor (S32K344 + iCE40 FPGA + Okika FPAA):** Architected a unified inline vehicle interceptor replacing separate vehicle-specific boards; selected an **NXP S32K344** (Automotive Ethernet PHY + 6x CAN-FD / 4x LIN), a **Lattice iCE40UP5K FPGA** (sized for deterministic LUT bit-timing of SENT, BEAN, and CXPI), and an **Okika FPAA** (analog/resistive switch spoofing) with a fail-safe **short-to-OEM (NC relay)** hardware bypass.
- **SENT Bus & Embedded Linux Voice HMI:** Adapted CAN/LIN control logic to **SAE J2716 SENT** with a custom interceptor PCB; built an Embedded Linux push-to-talk voice controller using **NXP VIT** and custom **Yocto**/**BitBake** recipes, refined production **Qt**/**QML** displays, authored a 129-test C verification suite, and modeled injection-molded/3D-printed enclosures in **Fusion 360** / **FreeCAD**.

Full-Stack Product & Systems Engineer — Independent Product & Open-Source Engineering | 2022 – Present

- Designed, built, and deployed full-stack web applications, developer tools, and automated data pipelines using **TypeScript, React, Next.js, Python, Rust, and PostgreSQL**/**SQLite**, owning product discovery, UI/UX, REST APIs, OAuth 2.0, and CI/CD.
- **Active Upstream Open-Source Contributor:** Contributing bug fixes and runtime improvements to **Tenstorrent (`tenstorrent/tt-metal`)** in the TT-NN / AI accelerator model execution stack and **T3 Code (`pingdotgg/t3code`)** in the TypeScript/React Agent Client Protocol (ACP) provider stack.

Flagship Open-Source Systems (github.com/mESmaC)

Electrace — Browser-Native Schematic & Multi-Layer PCB EDA Suite (TypeScript, React, Next.js, PostgreSQL, MCP, KiCad)

- Built a full-stack schematic capture and multi-layer PCB layout IDE with a custom SVG geometry engine, transitive power-rail propagation, deterministic ERC/DRC verification, native `.kicad_pcb` export, and a 20+ tool **Model Context Protocol (MCP)** server for autonomous AI circuit design.

Kadmium (Cd-48) — Formally Verified Distributed AI/ML Compute Fabric (100% Rust, Tokio, SafeTensors, Lean 4)

- Engineered a BOINC/CERN-inspired distributed Rust runtime that slices Llama/Mistral/Qwen SafeTensors layers across heterogeneous GPUs, backed by a zero-`sorry` **Lean 4** equivalence proof (`Kadmium.Native.runChain_eq_runLayers`) and bit-identical cut parity tests.

MCL4-COE & NCM — Markov-Chain Sparse Attention & 12 GB Consumer-GPU LLM Pretraining (PyTorch, CUDA, NVFP4, Hugging Face)

- Designed a custom LLM architecture pairing Markov-Chain Local-4 sparse attention, a Causal Encoder-Decoder MoE backbone, and a 9-head single-softmax Cognitive Operator Ensemble; built `CPUOffloadAdamW` and `NVFP4` 2:4 sparsity pipelines to pretrain and publish **2.43B** and **1.77B** models on a single 12 GB GPU.

Mirror & OIA — Freestanding x86_64 Rust OS Kernel & Formally Specified Provenance Engine (Rust, Assembly, Lean 4, QEMU)

- Built **Mirror**, a `no_std` `x86_64` SMP OS kernel in Rust (4-CPU APIC, per-process page tables, CoW `fork`, SMEP/SMAP/UMIP, VirtIO/AHCI/xHCI drivers, 35 syscalls, 156 QEMU fault-injection boots) and **OIA**, a multi-crate Rust public-intelligence suite with Lean 4 bounded-memory and cryptographic lineage proofs.